

Name: _____

Mark: _____ / 5

Mini-math Div 3/4: Wednesday, September 23, 2025 (5.8-5.12) - 10 minutes

1. (1 point) Consider the curve given by $x^2 + xy + y^2 = 7$. At the point $(1, 2)$, is the curve increasing or decreasing?
- A. Increasing
B. Decreasing
C. Neither
D. Cannot be determined

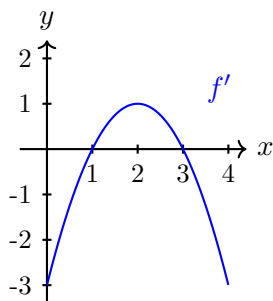
2. (1 point) The cost of producing a cylindrical can of radius r cm is

$$C(r) = \frac{1}{4}r^2 + \frac{27}{2r}, \quad r > 0.$$

At which value of r is the cost minimized?

- A. $\frac{3}{2}$
B. 2
C. 3
D. 9

3. (1 point) The graph of f' , the derivative of a twice-differentiable function f , is shown below. On which interval is f both increasing and concave down?



- A. $(1, 2)$
- B. $(2, 3)$
- C. $(3, 4)$
- D. $(0, 1)$

4. (1 point) A rectangle has its base on the x -axis and its upper two vertices on the parabola $y = 12 - x^2$. Which of the following expressions gives the area of the rectangle as a function of x , where (x, y) is one of the upper vertices?

- A. $A(x) = x(12 - x^2)$
- B. $A(x) = 2x(12 - x^2)$
- C. $A(x) = 2x(12 - x)^2$
- D. $A(x) = (2x)^2(12 - x^2)$

5. (1 point) The graph of f' , the derivative of a differentiable function f , is shown below. Which of the following is a possible graph of f ?

